

< Previous





Next >

Conditional Complexity

 Bookmark this page

Problem: Complexity does not only depend on the machine, but it also varies with the context!

n in the list of
nts in reverse order.
nowledge, Bill
s much less than if
n.

ay that the
ty of Bill Clinton
g that he is a former
t is the complexity of
is about $\log_2(k)$ bits),
: scan direction of the

$$sident') = \log_2(k) + 1$$

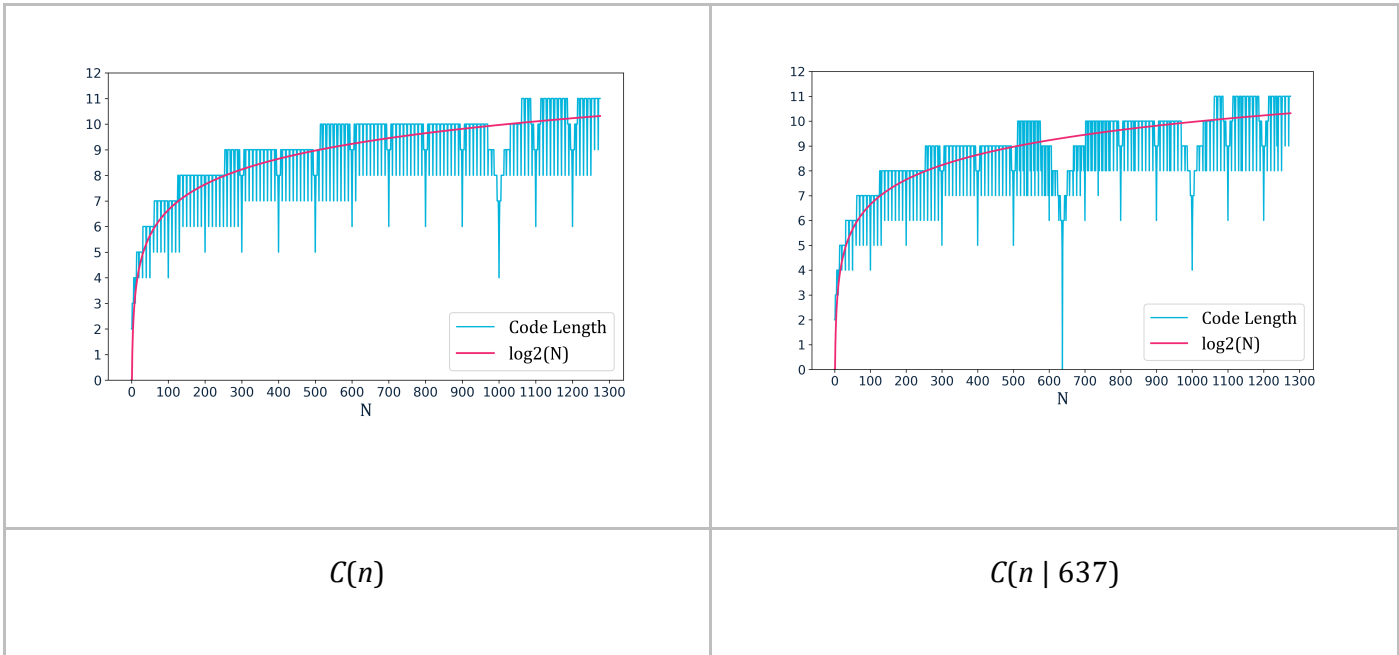
Conditional complexity

$$C_U(s_1 \mid s_2) = \min_p \{|p| : U(s_2, p) = s_1\}.$$



Caution: in the definition of conditional complexity, the notation $U(s_2, p)$ supposes that the machine U be given a way (such as a delimitation) to distinguish s_2 from p .

Conditional complexity offers a way of taking context into account. In a context in which the number 637 has just been mentioned, 637 itself and numbers close to it become automatically simple. The following images show $C(n)$ and $C(n \mid 637)$ using our coding scheme. We can verify that $C(637 \mid 637) = 0$, and that numbers around 637 have low complexity (hence the trough around 637 in the conditional complexity curve).



Now consider the number 2811. Using our coding scheme for integers, one can see that the description of 2811 requires no more than 11 bits. The program `CompactIntegerCoding.py` can be used to verify that 2811 is coded using 2800 as a reference. Let's first compute absolute complexities.

```
$ python
```

```
>>> import CompactIntegerCoding as CC
>>> CC.Coding(2811)
'00 1110 00 101'      # == '00' + <compact code of 2800> +
<compact code of 11>
>>> CC.Coding(2800)
'0 1110 00'          # == <compact code of 2800> (with heading
'0')
>>> CC.Coding(11)
'1 101'              # == <compact code of 11> (with heading
'1')
```

Conditional complexity can be used to verify that the description complexity of 2811 can be even smaller if 2800 is already in the context. In `our program`, conditional complexity is implemented by using a 3-bit long prefix: '100 ' for a positive shift, '101 ' for a negative shift.

```
$ python
>>> import CompactIntegerCoding as CC
>>> CC.ConditionalCoding(2811, 2800)
'100 1 101'          # note the heading '100' that marks
conditional computation; '1 101' = code for 11
>>> CC.Coding(2800)      # to put 2800 into memory (in
coded form)
'0 1110 00'
>>> CC.ConditionalDecoding('100 1 101')    # uses memory
content to retrieve the actual code for 2811
2811
```

For this program, 2811 is represented by 100 1 101 when 2800 is in short-term memory, instead of the longer code 00 1110 00 101 that contains the code for 2800 in explicit form (1110 00, which means '28 followed by 2 (= 00) zeros').

Conditional Complexity

1/1 point (ungraded)

Compute the conditional description length $C(2789|2800)$ (not counting space separators) given by the coding program `PositionalCoding.py`. What do you get?

- ☒ 7 bits
- ☐ 8 bits
- ☐ 11 bits
- ☐ 12 bits

✓
Answer
Correct:

```
$ python
>>> import CompactIntegerCoding as CC
>>> CC.ConditionalCoding(2789, 2800)
'101 1 101' # note the heading '101', where the last '1' marks negative
shift.
```

The length (in number of bits) of this conditional code is 7.

`Submit` You have used 1 of 1 attempt

Conditional vs Non Conditional

1/1 point (ungraded)

How does conditional complexity $C(x|y)$ compare to non-conditional complexity $C(x)$?

- ☒ Conditional complexity is always smaller (up to a constant)
- ☐ Conditional complexity is always larger (up to a constant)



Answer

Correct:

Programs that compute x when y is available for free cannot be longer than programs that compute x from scratch. We allow for an additional constant to organize the input switch from y to x on the machine.

Submit

You have used 1 of 1 attempt

Answers are displayed within the problem

[< Previous](#)

[Next >](#)



edX

- [About](#)
- [Affiliates](#)
- [edX for Business](#)
- [Open edX](#)
- [Careers](#)
- [News](#)

Legal

- [Terms of Service & Honor Code](#)
- [Privacy Policy](#)
- [Accessibility Policy](#)
- [Trademark Policy](#)
- [Sitemap](#)
- [Cookie Policy](#)
- [Your Privacy Choices](#)

Connect

- [Idea Hub](#)
- [Contact Us](#)
- [Help Center](#)
- [Security](#)
- [Media Kit](#)

